

Dynamic Pricing Engine

Project Overview

The Dynamic Pricing Engine is a pricing optimization platform inspired by e-commerce companies such as Amazon and Flipkart.

In large marketplaces, product prices are not fixed. Prices change continuously based on factors such as demand, inventory levels, competitor pricing, and seasonal trends.

The objective of this project is to build a system that automatically recommends the best selling price for a product using predefined business rules.

This project introduces students to real-world concepts such as pricing algorithms, rule engines, inventory management, analytics, and AI-assisted recommendations.

Problem Statement

An e-commerce company sells products through multiple channels.

If a product is selling too quickly, the company may want to increase the price.

If inventory is very high and sales are slow, the company may want to decrease the price.

If competitors are selling the same product at a lower price, the company may need to adjust pricing to remain competitive.

The challenge is to automatically determine the most suitable price for a product based on changing business conditions.

Project Objectives

- Build a pricing recommendation engine
- Manage products and inventory
- Simulate competitor pricing
- Implement dynamic pricing rules
- Generate pricing recommendations
- Visualize pricing trends and analytics

Core Features

Product Catalog

The system should allow:

- Add products
- View products
- Update products
- Manage base prices

Inventory Management

The system should track:

- Available stock
- Low stock products
- High stock products

Competitor Price Feed

The system should store competitor prices.

For simplicity, students can manually enter competitor prices instead of integrating real websites.

Example:

Product: Wireless Mouse

Our Price: ₹799

Competitor Price: ₹749

Pricing Engine

The pricing engine evaluates different factors and recommends a new selling price.

Analytics Dashboard

The dashboard should display:

- Product prices
- Price changes
- Inventory status
- Pricing recommendations

Pricing Factors

Demand

If demand is high:

- Increase price slightly

If demand is low:

- Decrease price slightly

Inventory

If inventory is low:

- Increase price

If inventory is high:

- Decrease price

Competitor Price

If competitors are significantly cheaper:

- Recommend a lower price

If competitors are more expensive:

- Recommend a higher price

Time-Based Rules

Example:

- Weekend Sale
- Festival Discount
- End of Month Offers

Pricing Logic

Students should start with simple rule-based pricing.

Example Rules:

Rule 1

If demand is high and inventory is low

Increase price by 10%

Rule 2

If demand is low and inventory is high

Decrease price by 10%

Rule 3

If competitor price is lower

Reduce price by 5%

Rule 4

If competitor price is higher

Increase price by 3%

The final recommended price is calculated after applying applicable rules.

AI Feature

AI should not directly set product prices.

Instead, AI should explain pricing decisions.

Example:

Input:

Product: Wireless Mouse

Current Price: ₹799

Demand: High

Inventory: Low

Competitor Price: ₹850

Output:

"The recommended price is ₹879 because demand is high, inventory is limited, and competitor prices are currently higher."

This makes the system more explainable and realistic.

Functional Requirements

Product Management

The system should allow:

- Add products
- Edit products
- Delete products
- View product catalog

Inventory Management

The system should allow:

- Update stock
- View inventory levels
- Monitor stock status

Competitor Pricing

The system should allow:

- Add competitor prices
- Update competitor prices
- Compare prices

Pricing Engine

The system should allow:

- Calculate recommended price
- Apply pricing rules
- Store pricing history

Database Design

Products Collection

Fields:

- id
- productName
- category
- currentPrice
- createdAt

Inventory Collection

Fields:

- id
- productId
- availableQuantity
- updatedAt

Competitor Prices Collection

Fields:

- id
- productId
- competitorName
- competitorPrice
- updatedAt

Pricing History Collection

Fields:

- id
- productId
- oldPrice
- recommendedPrice
- reason
- createdAt

Backend APIs

Product APIs

POST /products

GET /products

PUT /products/:id

DELETE /products/:id

Inventory APIs

POST /inventory

GET /inventory

PUT /inventory/:id

Competitor APIs

POST /competitor-prices

GET /competitor-prices

Pricing APIs

POST /calculate-price

GET /pricing-history

Frontend Pages

Dashboard

Display:

- Total Products
- Low Stock Products
- Price Recommendations
- Recent Price Changes

Product Management Page

Display:

- Product List
- Product Prices
- Categories

Skill Turtle

Inventory Management Page

Display:

- Stock Levels
- Inventory Status

Competitor Pricing Page

Display:

- Competitor Prices
- Price Comparison

Pricing Recommendation Page

Display:

- Current Price
- Recommended Price
- Pricing Factors
- Explanation

This should be the primary demo page during project presentations.

Application Flow

1. Admin adds products.
2. Inventory levels are updated.
3. Competitor prices are entered.
4. Pricing engine evaluates business rules.
5. Recommended price is generated.
6. Recommendation is stored.
7. AI generates explanation.
8. Results are displayed on the dashboard.

Technology Stack

Frontend

- React.js
- Material UI
- Axios

Backend

- Node.js
- Express.js

Database

- MongoDB

AI Integration

- OpenAI API or Gemini API

Deployment

- Frontend: Vercel
- Backend: Render
- Database: MongoDB Atlas

Team Responsibilities

Student 1 – Frontend Developer

Responsibilities:

- Dashboard
- Product Management Pages
- Inventory Pages
- Pricing Recommendation Screens

Student 2 – Backend Developer

Responsibilities:

- APIs
- Database Design
- Business Logic
- Authentication

Student 3 – Pricing and AI Developer

Responsibilities:

- Pricing Engine

- Rule Engine
- AI Explanation Module
- Testing and Documentation

Two-Week Development Plan

Days 1–2

Planning and Design

- Finalize requirements
- Create wireframes
- Design database schema

Days 3–5

Backend Development

- Product APIs
- Inventory APIs
- Pricing APIs

Days 6–8

Frontend Development

- Dashboard
- Product Pages
- Inventory Pages
- Pricing Screens

Days 9–10

Pricing Engine

- Demand Rules
- Inventory Rules
- Competitor Pricing Rules

Days 11–12

Analytics

- Pricing History

- Recommendation Dashboard

Day 13

AI Integration

- Pricing Explanation Generation

Day 14

Testing and Deployment

- Bug Fixes
- Deployment
- Documentation

Resume Description

Developed a Dynamic Pricing Engine inspired by modern e-commerce platforms.

Built a rule-based pricing system that generates optimal product pricing recommendations using demand, inventory levels, competitor pricing, and business rules.

Implemented inventory management, competitor price tracking, and pricing analytics using React, Node.js, Express, and MongoDB.

Integrated AI-generated explanations to provide transparent and understandable pricing recommendations.

Interview Discussion Points

Why was this project built?

Large e-commerce companies continuously adjust product prices based on market conditions. This project demonstrates how pricing decisions can be automated using business rules and analytics.

How does the pricing engine work?

The engine evaluates demand, inventory levels, competitor prices, and predefined business rules. Based on these factors, it recommends an updated product price.

Why was AI used?

AI was used to explain pricing recommendations. The actual pricing decision remains rule-based and easy to understand.

How can this system be scaled?

Future improvements may include:

- Real-time competitor price scraping
- Machine learning-based price prediction
- Customer demand forecasting
- Seasonal pricing strategies
- Automated promotion management

Why Recruiters Like This Project

This project demonstrates understanding of:

- Business rule engines
- Pricing algorithms
- Inventory management
- Analytics dashboards
- Full-stack development
- AI-assisted decision systems

These concepts are commonly used in e-commerce, retail technology, and marketplace platforms.